

3. Exercise: Training a Model

name(s): _____ points: _____ (max. 24)

1. Part Model Training

24 points

Train a model based on the data you collected in Exercise 2. It should be a deep learning model using TensorFlow. After training the model, make sure the conversion to a TFLite model works. In the next lesson, you will deploy the model to the Raspberry Pi and evaluate its performance on the device.

- (a) As mentioned in the last lesson you should have collected enough data to train the model. You need to collect samples for all three valid classes and the garbage class. We would recommend to have **at least 60 samples per class**, but more is always better.
- (b) Train a deep learning model using TensorFlow. We would recommend to specify your own model architecture.
- (c) Pay attention to creating an appropriate training, validation and test split.
- (d) When selecting the layers of the model, make sure that the model is convertible to TFLite. You should check this by attempting a TFLite conversion.
- (e) Train the model and evaluate its performance on the test set. You should achieve an accuracy that is at least 70% on the test set.
- (f) If you have problems achieving this accuracy, try to adjust the model architecture and hyperparameters or use augmentation techniques to increase the size of your dataset. Optionally you can also collect more data.
- (g) After training the model, convert your model to TFLite and check if the conversion was successful.
- (h) The entire exercise can be implemented in a devcontainer and follow the same pattern as the model training and conversion in the lesson.
- (i) You need not document your code other than providing comments in the code itself.
- (j) The code must be your own work. You can use online resources to understand how to implement the model, but you must not use generative AI tools to write the entire code for you.
- (k) You must be able to explain your code and design decisions in the lesson. Failure to know your code and design decisions will lead to negative grading of the exercise.